Bayesian Information Borrowing in Basket Trials: A Simulation Study Motivated by the FDA's 2026 Draft Guidance

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Abstract

The U.S. Food and Drug Administration released a draft guidance in 2026 that recognizes Bayesian information borrowing as an important tool for clinical trials. Motivated by this regulatory development, this study investigates how information borrowing performs in basket trials as subgroup heterogeneity increases. Through a simulation study involving four baskets, 30 patients per basket, six scenarios, three analytical strategies—independent analysis, complete pooling, and hierarchical Bayesian modeling—were compared using estimation accuracy, interval performance, and decision outcomes. The results show that hierarchical Bayesian models provide a practical balance between efficiency and robustness, achieving efficiency gains when subgroups are similar while limiting inappropriate borrowing when meaningful differences exist. These findings help clarify the practical conditions under which information borrowing may be responsibly applied in small-population clinical trials.

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1 Introduction

For the past few decades, frequentist methods have dominated the regulatory framework for clinical trials of drugs and biological products in the United States. The null hypothesis significance testing framework, characterized by fixed sample sizes and pre-specified alpha-allocation plans, provides reliable error control under standard conditions (Berry, 2006). Yet this framework encounters an increasingly well-recognized limitation: when the disease under study is rare, or when the population of interest is a vulnerable subgroup such as pediatric patients, the sample size required by a conventional frequentist design can exceed the total number of patients available globally. This deficit is a structural barrier to evidence generation rather than a mere operational inconvenience.

To address this challenge, Bayesian methods offer an alternative inferential logic. Rather than treating each trial as an isolated experiment that must stand on its own evidence, the Bayesian paradigm formally incorporates secondary or external data sources—such as historical controls, real-world registries, or parallel subgroups within a concurrent master protocol—through the mechanism of prior distributions. This practice, often referred to as information borrowing, mathematically updates existing knowledge with concurrent trial observations (Zhou & Ji, 2024). It is the direct operational consequence of the Bayesian principle that inference should update prior knowledge with observed data, and it holds particular promise for settings where data are intrinsically scarce.

This Bayesian approach is not merely theoretical; it has already been applied in clinical trials relevant to drug development. One notable example is the phase 2 trial of BI 1015550 for idiopathic pulmonary fibrosis (Richeldi et al., 2022). In this study, investigators used a Bayesian model to analyze change in forced vital capacity, with analyses reported according to background antifibrotic use. The Bayesian framework allowed the investigators to express treatment evidence directly in terms of posterior probabilities, which is especially useful when sample sizes are limited. This case illustrates how Bayesian methods can support clinical inference in settings where conventional evidence generation is constrained.

As highlighted by such applications, the practical need for sample efficiency eventually led to a major regulatory shift. The regulatory landscape saw another big update in January 2026, when the U.S. Food and Drug Administration (FDA) released a draft guidance titled *Use of Bayesian Methodology in Clinical Trials of Drug and Biological Products* (U.S. Food and Drug Administration [FDA], 2026). While the agency had previously issued relevant guidance for medical devices and Bayesian approaches had already seen widespread adoption over the past decade, this document establishes a dedicated framework for Bayesian methods targeting drugs and biological products. It highlights information borrowing as a key application that merits rigorous evaluation. This caution is important because the same mechanism that improves efficiency may also introduce bias when the borrowed information is not sufficiently comparable to the target subgroup. Therefore, the guidance does not give unconditional approval. It requires sponsors to prove that external data are comparable to the trial population, and it insists that prior specifications be justified through sensitivity analyses. In a JAMA Perspective article, Lee et al. (2026) characterized the guidance as "long-awaited" and argued that Bayesian methods are especially valuable in rare diseases, pediatrics, and oncology—precisely the domains where the sample-size bottleneck is most severe. These updated regulatory guidelines highlight the importance of biostatistics in modern clinical research. Beyond mere data processing, biostatisticians are tasked with quantifying complex uncertainties and balancing analytical efficiency against patient safety and regulatory standards. This dual duty is especially critical for new trial designs that rely on responsibly sharing data across diverse patient sub-populations.

Motivated by this regulatory development, this report investigates a central question: how does the performance of Bayesian information borrowing change as subgroup heterogeneity increases? Through a simulation study comparing independent analysis, complete pooling, and hierarchical Bayesian modeling, the report explores the practical conditions under which information borrowing is beneficial and the circumstances under which it may become problematic. In doing so, the study illustrates the role of biostatistics in balancing efficiency, reliability, and patient protection under the current FDA framework.

2 Related Work

2.1 The Evolution of Bayesian Methods in Clinical Trials

Bayesian statistics has gradually moved from a largely theoretical alternative to a practical framework for clinical trial design and analysis. This transition reflects the decision-oriented nature of clinical research, where investigators must often evaluate accumulating evidence while considering uncertainty, prior information, and future trial decisions. Unlike the frequentist approach, which relies on long-run frequencies under a fixed null hypothesis, the Bayesian paradigm naturally accommodates the cumulative updating of evidence as data accrue (Lee & Chu, 2012).

As articulated by Berry (2006), Bayesian methods offer several distinct practical advantages: Bayesian methods allow for the continuous incorporation of accumulating interim data, yield direct probability statements regarding treatment effects, and support flexible adaptive decision-making pathways. Early foundations in regulatory Bayesian principles were established by Spiegelhalter et al. (2004), who formalized how subjective and objective prior components could be synthesized in clinical research. More recently, Berry (2025) reviewed thirty years of adaptive trials and identified the capacity for real-time observation of cumulative data and dynamic trial modification as the Bayesian approach's most consequential contribution.

Over time, the methodological toolkit has expanded greatly. Early Bayesian clinical trial designs focused on single-arm phase II studies and dose-finding algorithms. By the 2000s and 2010s, the landscape had broadened to encompass platform trials, basket trials, and umbrella designs—complex trial architectures that simultaneously evaluate multiple treatments, multiple disease subtypes, or both (Park et al., 2019; Woodcock & LaVange, 2017). These designs are natural hosts for Bayesian methods because their very structure embeds a borrowing problem: subgroups within the trial share a common investigative framework, and failing to exploit this shared structure is statistically inefficient.

2.2 Regulatory Acceptance: From Medical Devices to Drugs and Biologics

The FDA's engagement with Bayesian methods followed a two-stage trajectory. In 2010, the agency issued its formal "Guidance for the Use of Bayesian Statistics in Medical Device Clinical Trials" (U.S. Food and Drug Administration, 2010), providing a formal statistical framework for Bayesian designs in the device sector. This guidance proved consequential: Bayesian adaptive designs became a standard option for device trials, and the document established regulatory precedents—on prior specification, sensitivity analysis, and the role of simulation in design justification—that shaped the broader methodological conversation.

For drugs and biologics, however, an equivalent framework remained absent for over fifteen years. Sponsors sometimes applied Bayesian methods for exploratory or supplementary analyses, yet there was no clear pathway to use Bayesian approaches for primary analyses in pivotal clinical trials. The January 2026 draft guidance (FDA, 2026) fills the gap. Its primary focus, as the document itself states, is "the use of Bayesian methods to support primary inference in clinical trials intended to support the demonstration of effectiveness or safety." Crucially, the guidance identifies information borrowing across historical controls or parallel synchronous subgroups as a key area requiring rigorous pre-trial validation.

Medical and biostatistical researchers view the guidance as a conceptual change rather than just a procedural update (Gelman et al., 2026). Where randomized evidence is scarce, Bayesian methods use external data and measure uncertainty in a more flexible and transparent way than conventional frequentist analysis.

2.3 Information Borrowing Methods

Several methodological frameworks have been proposed to support Bayesian information borrowing in clinical research. Information borrowing in biostatistics can be broadly divided into two types: longitudinal borrowing (using past historical data) and horizontal borrowing (sharing data across current subgroups) (Chen & Lee, 2018; Zhou & Ji, 2024).

For historical borrowing, commonly used methods include power priors, mixture priors, and commensurate priors (Hobbs et al., 2011; Ibrahim & Chen, 2000). These approaches regulate borrowing strength according to the similarity between historical and current evidence and have been widely applied in settings involving external control information.

In contrast, basket and platform trials often require borrowing across concurrently enrolled patient subgroups rather than across historical studies (Thall et al., 2003). In such settings, the hierarchical Bayesian model (HBM) has emerged as a dominant framework. Here, the treatment effects of different subgroups are assumed to come from a single shared population. The borrowing mechanism is entirely data-driven, governed by the posterior distribution of the between-group variance parameter (τ^2).

For a multi-cohort basket trial simulation without any external historical data, the HBM is the natural choice. If the subgroups look similar, τ^2 becomes small, and the model pools the data heavily. If the subgroups look different, τ^2 grows large, reducing the amount of borrowing and making the subgroup-specific estimates behave more similarly to an independent analysis. HBM allows the concurrent data themselves to determine the borrowing weights. However, this data-driven flexibility also introduces unique operational risks under certain conditions, which requires careful examination.

2.4 Limitations and Implementation Challenges of the HBM

To understand the operational risks mentioned above, it is necessary to examine how the HBM performs in practical settings. In particular, applied biostatisticians have highlighted major problems when the model is used in small trials (Gelman, 2006).

The first challenge is the small number of subgroups. In a typical multi-cohort basket trial, the number of cohorts is often limited (e.g., fewer than 10). With so little data, the model struggles to accurately estimate the between-group variance parameter (τ^2). As a result, the final statistical analysis depends too heavily on the prior distribution that researchers subjectively choose for τ^2 (Neuenschwander et al., 2016).

This estimation problem becomes dangerous when the true treatment effects across subgroups are highly unequal (Roche et al., 2021). For example, consider a trial with four subgroups, three of which have no treatment effect while only one shows a strong effect. The HBM can mistakenly "leak" the success of the active group into the inactive ones. This data sharing causes a serious issue: it inflates the Type I error rate, meaning inactive groups are falsely labeled as successful. This directly violates regulatory standards for statistical evidence and raises serious ethical concerns regarding patient safety.

3 Motivation and Problem Formulation

3.1 The Sample-Size Barrier in Stratified Populations

Basket trials have become an important design response to increasingly stratified patient populations, where a common therapeutic hypothesis is evaluated across multiple subgroups. However, because evidence within each subgroup can be sparse, these designs also create statistical challenges for borrowing, subgroup-specific inference, and decision-making (Hobbs et al., 2022). In this setting, when sample sizes are strictly limited—such as the $N_k = 30$ per cohort constraint evaluated in this study—individual subgroup analyses often lack the power to support definitive inference. The result is a design problem in which clinically important therapeutic questions may remain difficult to evaluate using conventional methods.

This sample-size barrier is a defining challenge in modern oncology and orphan diseases. Between 2010 and 2020, the FDA's Office of Orphan Products Development designated over 500 new orphan drugs, many of which target patient populations highly fragmented into small molecular subpopulations (Miller et al., 2021). Pediatric trials face a similar ethical and practical imperative: researchers must minimize children's exposure to experimental therapies while navigating extremely small sample sizes that render standard confirmatory trials impractical. In this context, rigorous biostatistical methodology becomes indispensable for generating reliable clinical evidence.

3.2 Horizontal Borrowing as a Structural Solution

Multi-cohort basket trials utilizing Bayesian horizontal borrowing offer a mathematical solution to this enrollment bottleneck. By sharing information across concurrent subgroups, investigators may reduce the required per-basket sample size by increasing each cohort's effective sample size, while maintaining statistical rigor when borrowing assumptions are adequately justified. The FDA 2026 guidance acknowledges this logic, stating that Bayesian methods can "make better use of available data" and "conduct more efficient clinical trials" (FDA, 2026).

Yet, as shown in the previous section, this information sharing is not a free lunch. The central tension lies in the risk of false borrowing when the underlying subgroups are highly heterogeneous. While aggressive information borrowing minimizes Mean Squared Error and stabilizes variance when cohorts are homogeneous, it simultaneously exposes the trial to severe Type I error inflation and estimation bias under localized efficacy profiles.

From a protocol design perspective, this creates a dilemma in practice: an investigator cannot maximize borrowing efficiency without inherently increasing the risk of posterior contamination. The regulatory and methodological challenge is to map this risk-reward profile, establishing the operational boundaries where the adaptive shrinkage of the HBM remains safe, reliable, and ethically justifiable.

3.3 The Practical Limits of Information Borrowing

This operational tension between efficiency gains and false borrowing leads directly to our core research question: How does the relative performance of hierarchical Bayesian borrowing change across different patterns and degrees of subgroup heterogeneity? Specifically, we want to understand the situations in which the HBM provides a balance between efficiency and robustness relative to both independent and fully pooled analyses. To answer this question, we formalize a comprehensive simulation study that subjects these competing strategies to diverse configuration profiles—ranging from complete subgroup homogeneity to highly localized efficacy outliers—and quantifies their behavior through key frequentist and decision-theoretic operating characteristics.

4 Methods and Simulation Framework

4.1 Simulation Objectives

The primary objective of this simulation study was not simply to compare competing statistical models, but to explore the practical limits of Bayesian information borrowing under varying degrees of subgroup heterogeneity.

Specifically, three research questions were investigated:

1. Efficiency under exchangeability: When treatment effects are similar across subgroups, can hierarchical Bayesian models recover the efficiency gains achieved by complete pooling?

2. Adaptation to heterogeneity: As subgroup response rates become increasingly heterogeneous, does the hierarchical model automatically reduce borrowing and maintain reliable estimation performance?

3. Protection against false borrowing: Under asymmetric scenarios where only a subset of cohorts differs substantially from the remainder, can the hierarchical model prevent inappropriate information sharing that may lead to biased inference or incorrect regulatory decisions?

To address these questions, a simulation framework was constructed with varying degrees and structures of between-subgroup heterogeneity while holding all other design characteristics fixed.

4.2 Data Generating Process and Evaluation Scenarios

Let us consider a basket trial with $K = 4$ distinct subgroups, each evaluating a binary efficacy outcome (response or non-response). For each simulated trial $m \in \{1, 2, \ldots, M\}$ ($M = 1000$), the primary endpoint for subgroup $k \in \{1, \ldots, K\}$ is modeled as an independent Binomial process:

$$y_{m,k} \sim \text{Binomial}(N_k, \theta_{m,k}) \tag{1}$$

where $\theta_{m,k} = \theta_k$ denotes the true response rate of subgroup k; $N_k = 30$ represents the fixed sample size per cohort, yielding a total of 120 patients per simulated trial. The historical control threshold is fixed at $\theta_0 = 0.20$ across all evaluations. The choice of four baskets and 30 patients per basket was intended to mimic the small-sample settings commonly encountered in rare-disease and pediatric trials, where information borrowing is most likely to be considered.

Six scenarios were specified covering varying degrees of between-subgroup heterogeneity. The first three scenarios evaluated the effect of increasing levels of global heterogeneity:

1. Homogeneous (All Active): all subgroups share a common treatment effect.
2. Moderate Heterogeneity: subgroup effects differ modestly while remaining clinically similar.
3. Strong Heterogeneity: subgroup effects vary substantially, challenging the exchangeability assumption.

The remaining three scenarios examined asymmetric configurations that are particularly relevant to regulatory evaluation:

4. Single Extreme Dropout: one subgroup performs substantially worse than the others.
5. Global Null (All Inactive): all subgroups are inactive, providing a benchmark for evaluating false positive behavior.
6. Single Super Winner: only one subgroup exhibits strong efficacy while all others remain inactive, representing the most severe false-borrowing stress test.

4.3 Comparison Strategies

This study evaluates and compares three distinct analytical strategies applied to the same simulated datasets:

Table 1: True response rates across different scenarios and baskets

Scenario	Basket 1	Basket 2	Basket 3	Basket 4
S1: Homogeneous (All Active)	0.35	0.35	0.35	0.35
S2: Moderate Heterogeneity	0.25	0.30	0.35	0.40
S3: Strong Heterogeneity	0.15	0.25	0.45	0.55
S4: Single Extreme Dropout	0.10	0.30	0.30	0.30
S5: Global Null (All Inactive)	0.15	0.15	0.15	0.15
S6: Single Super Winner	0.55	0.15	0.15	0.15

1. Independent Analysis. Each of the four subgroups is analyzed separately with a flat, non-informative Beta$(1, 1)$ prior. No information is shared across cohorts, serving as the unbiased but statistically inefficient baseline.

2. Complete Pooling. All subgroups are aggregated into a single unified population with a Beta$(1, 1)$ prior. This represents the extreme baseline of unadjusted borrowing under full exchangeability assumptions.

3. Hierarchical Bayesian Model. The adaptive information borrowing framework under evaluation, specified via the non-centered parameterization detailed below.

4.4 Hierarchical Bayesian Model Specification

To avoid MCMC convergence issues in small samples ($N_k = 30$), I fitted the hierarchical model on the logit scale with a non-centered parameterization:

$$\text{logit}(\theta_{m,k}) = \mu_m + \eta_{m,k} \cdot \tau_m, \tag{2}$$

$$\eta_{m,k} \sim \mathcal{N}(0, 1), \tag{3}$$

$$\mu_m \sim \mathcal{N}(0, 2^2), \tag{4}$$

$$\tau_m \sim \text{Half-Normal}(1). \tag{5}$$

The between-group standard deviation τ_m is the core hyperparameter governing the borrowing mechanism. When the posterior of τ_m concentrates near zero, the model shrinks subgroup estimates toward the common mean μ_m, approximating complete pooling. Conversely, when the posterior of τ_m shifts toward larger values, the random effects absorb the between-subgroup variation, and the model adaptively transitions toward an independent analysis. The Half-Normal(1) hyperprior on τ_m serves as a weakly informative default that places its primary prior mass on small to moderate heterogeneity while allowing strong data-driven evidence to drive the posterior upward.

All models were compiled and executed using `PyMC` (v5) utilizing the JAX-backed `numpyro` No-U-Turn Sampler engine. To optimize computational throughput and ensure the feasibility of executing extensive sampling on a personal laptop, the computation was vectorized. For each individual trial, 1,000 posterior inference draws were collected following 1,000 tuning iterations across 4 parallel chains. Sampling warnings and divergent transitions were monitored during posterior computation. Replicates with severe sampling warnings were examined to ensure that the main conclusions were not driven by numerical failures.

4.5 Evaluation Metrics

Let $\hat{\theta}_{m,k}$ denote the posterior mean estimate for subgroup k in simulated trial m, and let $[L_{m,k}, U_{m,k}]$ represent its corresponding equal-tailed 95% posterior credible interval. To rigorously evaluate and compare the frequentist operating characteristics of the three analytical strategies, I formalized the evaluation metrics as follows:

- Overall Bias: This metric measures the average signed estimation error across all simulated trials and all subgroups:

$$\text{Overall Bias} = \frac{1}{M \cdot K} \sum_{m=1}^{M} \sum_{k=1}^{K} (\hat{\theta}_{m,k} - \theta_k). \tag{6}$$

  A positive value indicates systematic overestimation, whereas a negative value indicates systematic underestimation. Because positive and negative subgroup-level biases may cancel each other out, Overall Bias should be interpreted together with other metrics.

- Overall Mean Squared Error (Overall MSE): This metric quantifies the total estimation accuracy by pooling the squared errors across all M simulations and all K subgroups:

$$\text{Overall MSE} = \frac{1}{M \cdot K} \sum_{m=1}^{M} \sum_{k=1}^{K} (\hat{\theta}_{m,k} - \theta_k)^2. \tag{7}$$

- Maximum Absolute Deviation (MaxAD): To capture the most extreme localized estimation error observed under each scenario and analytical strategy, I evaluated the maximum absolute deviation across all simulated trials and all subgroups:

$$\text{MaxAD} = \max_{1 \leq m \leq M,\ 1 \leq k \leq K} \left| \hat{\theta}_{m,k} - \theta_k \right|. \tag{8}$$

  This metric is useful for identifying the worst-case estimation error that may occur under a given borrowing strategy.

- Empirical Coverage Rate: The empirical coverage rate was defined as the proportion of subgroup-level 95% posterior credible intervals that contained the corresponding true response rate:

$$\text{Coverage Rate} = \frac{1}{M \cdot K} \sum_{m=1}^{M} \sum_{k=1}^{K} \mathbb{I}\left(L_{m,k} \leq \theta_k \leq U_{m,k}\right), \tag{9}$$

  where $\mathbb{I}(\cdot)$ is the indicator function. Although these intervals are Bayesian credible intervals, their repeated-sampling coverage was evaluated as a frequentist operating characteristic across simulated trials.

- Probability of Correct Decision (PCD): A subgroup was declared successful if the posterior probability of exceeding the historical control threshold $\theta_0 = 0.20$ was greater than 95%, namely

$$\Pr(\theta_{m,k} > 0.20 \mid \text{data}) > 0.95. \tag{10}$$

  Let $D_{m,k}$ denote the model-based success decision and let T_k denote the true subgroup status, where $T_k = 1$ if $\theta_k > 0.20$ and $T_k = 0$ otherwise. PCD was defined as

$$\text{PCD} = \frac{1}{M \cdot K} \sum_{m=1}^{M} \sum_{k=1}^{K} \mathbb{I}\left(D_{m,k} = T_k\right). \tag{11}$$

For active subgroups, this metric corresponds to empirical power; for inactive subgroups, it corresponds to $1 -$ Type I error rate. Therefore, scenario-level PCD is used as a compact summary measure, while coverage and scenario-specific interpretation are also considered when assessing false borrowing risk.

5 Results

Table 2 summarizes the operating characteristics of the three analytical strategies across six scenarios. Rather than focusing on individual scenarios separately, the results reveal several broader patterns regarding when information borrowing is beneficial, when it becomes harmful, and how hierarchical borrowing adapts to changing levels of subgroup heterogeneity.

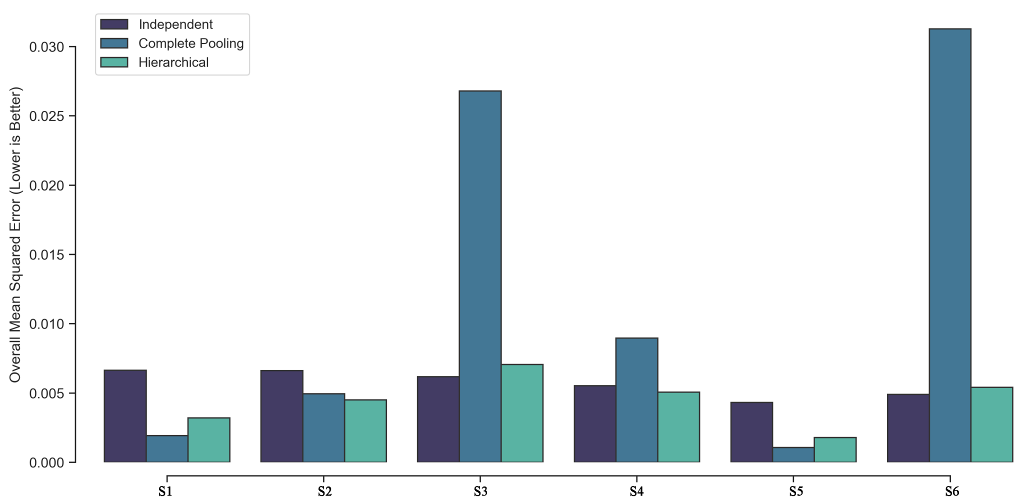

Figure 1: Mean Squared Error Comparison Across Simulation Scenarios. Lower values indicate improved estimation accuracy.

Table 2: Summary of Simulation Operating Characteristics Across Analytical Strategies.

Scenario	Strategy	Overall Bias	MaxAD	Overall MSE	Coverage Rate	PCD
S1: Homogeneous (All Active)	Complete Pooling	0.0016	0.1697	0.0019	0.9480	0.9800
	Hierarchical	0.0004	0.2478	0.0032	0.9790	0.8038
	Independent	0.0085	0.3062	0.0067	0.9475	0.6325
S2: Moderate Heterogeneity	Complete Pooling	0.0033	0.2090	0.0050	0.7395	0.9180
	Hierarchical	0.0020	0.2965	0.0045	0.9540	0.6292
	Independent	0.0114	0.3375	0.0066	0.9510	0.5162
S3: Strong Heterogeneity	Complete Pooling	0.0036	0.3500	0.0268	0.1800	0.7425
	Hierarchical	0.0025	0.2844	0.0071	0.9332	0.7832
	Independent	0.0104	0.2938	0.0062	0.9518	0.7808
S4: Single Extreme Dropout	Complete Pooling	0.0036	0.2852	0.0090	0.5740	0.4315
	Hierarchical	0.0019	0.2273	0.0051	0.9245	0.4740
	Independent	0.0152	0.2625	0.0055	0.9380	0.5515
S5: Global Null (All Inactive)	Complete Pooling	0.0056	0.1123	0.0011	0.9470	1.0000
	Hierarchical	0.0035	0.1807	0.0018	0.9768	0.9990
	Independent	0.0218	0.2562	0.0043	0.9622	0.9878
S6: Single Super Winner	Complete Pooling	0.0056	0.4025	0.0313	0.0982	0.5705
	Hierarchical	0.0040	0.3106	0.0054	0.9508	0.9892
	Independent	0.0170	0.2938	0.0049	0.9615	0.9908

MaxAD = Maximum Absolute Deviation; PCD = Probability of Correct Decision. All metrics were calculated from $M = 1000$ simulated trials per scenario. For the hierarchical Bayesian model, posterior inference was performed using Markov Chain Monte Carlo.

5.1 Efficiency Gains Under Exchangeability

When subgroup response rates were truly homogeneous (Scenario 1), information borrowing produced efficiency gains. Complete pooling achieved the lowest MSE (0.0019) and the highest probability of correct decision (PCD = 0.9800), reflecting the fact that the exchangeability assumption was perfectly satisfied. Under these ideal conditions, combining all observations into a single analysis maximized effective sample size and minimized estimation variance.

The hierarchical model also benefited from borrowing, reducing MSE by more than 50% relative to independent analysis (0.0032 versus 0.0067). However, its decision performance remained lower than that of complete pooling. This result is consistent with the model's adaptive structure: rather than enforcing full exchangeability, hierarchical borrowing retains uncertainty about between-subgroup variation and therefore behaves more conservatively even when the true subgroup effects are identical.

These findings suggest that when treatment effects are genuinely similar across cohorts, borrowing can noticeably improve statistical efficiency. In such settings, the hierarchical model successfully captures much of the benefit of complete pooling while retaining the flexibility to accommodate potential heterogeneity.

5.2 Performance Across Increasing Heterogeneity

As subgroup response rates became increasingly heterogeneous, the relative performance of the competing strategies changed markedly.

Under moderate heterogeneity (Scenario 2), the hierarchical model achieved the lowest overall MSE (0.0045), outperforming both complete pooling (0.0050) and independent analysis (0.0066). At the same time, coverage remained close to the nominal 95% level (95.4%), whereas complete pooling exhibited substantial under-coverage (73.9%). This scenario represents the setting in which hierarchical borrowing provided its greatest practical advantage: enough similarity existed to justify partial information sharing, yet sufficient differences remained that complete pooling became overly aggressive.

The contrast became even more apparent under strong heterogeneity (Scenario 3). Complete pooling experienced a dramatic deterioration in performance, with MSE nearly four times larger than the other models, and coverage collapsing to only 18.0%. In comparison, both the hierarchical and independent approaches maintained stable estimation accuracy and greatly improved interval calibration (see Figure 1).

Taken together, these results indicate that hierarchical borrowing derives its primary value not from dominating all competing methods, but from providing a robust compromise when the degree of subgroup exchangeability is uncertain. While complete pooling performs best under perfect homogeneity and independent analysis becomes preferable under extreme heterogeneity, the hierarchical model remains competitive across both settings and performs particularly well in the intermediate regime, which is likely to arise in many real-world basket trials.

5.3 Protection Against False Borrowing

The asymmetric scenarios provide a more stringent evaluation of borrowing behavior because they mimic situations in which only a subset of cohorts differs substantially from the remainder.

In Scenario 4, three cohorts shared a common response rate while a single subgroup exhibited markedly poorer outcomes. Complete pooling suffered severe performance degradation, with coverage falling to 57.4%, reflecting substantial bias caused by inappropriate borrowing across non-exchangeable subgroups. By contrast, the hierarchical model maintained reasonably good coverage (92.5%) and achieved slightly lower overall MSE than independent analysis (0.0051 versus 0.0055). However, its probability of correct decision was lower than that of independent analysis (0.4740 versus 0.5515). This suggests that partial borrowing improved estimation accuracy on average, but did not necessarily improve decision performance under this asymmetric dropout pattern. Overall, the hierarchical model partially insulated the atypical subgroup from excessive contamination, although some decision-level cost remained.

Scenario 6 provides a particularly stringent evaluation of borrowing behavior, where only a single subgroup demonstrated meaningful treatment efficacy while all remaining cohorts were inactive. This configuration directly evaluates the risk of false borrowing, a central regulatory concern in basket trial design. Under complete pooling, estimation performance collapsed: coverage fell to only 9.8%, indicating severe distortion of subgroup-specific inference. In contrast, the hierarchical model maintained coverage at 95.1%, closely matching the performance of independent analysis (96.2%), while maintaining low overall bias.

The subgroup shrinkage patterns shown in Figure 2 provide further insight into this behavior. Rather than blindly sharing information across all cohorts, the hierarchical model weakened borrowing as evidence of non-exchangeability accumulated. This adaptive response helped prevent the highly active subgroup from inflating efficacy estimates in inactive cohorts, thereby reducing the risk of erroneous regulatory conclusions.

5.4 Operating Characteristics Under the Global Null

Scenario 5 evaluated model behavior when all subgroups were inactive. This setting is particularly important because any borrowing strategy must avoid generating spurious evidence of efficacy when no true treatment effect exists.

All three methods performed reasonably well under the global null. Coverage remained close to nominal levels, ranging from 94.7% to 97.7%. Decision performance was also highly stable across methods. Notably, the hierarchical model did not exhibit evidence of systematic over-borrowing despite the strong similarity among cohorts. Instead, its operating characteristics remained comparable to those of independent analysis while benefiting from modest gains in estimation precision.

These findings suggest that hierarchical borrowing does not inherently create false signals when subgroup outcomes are uniformly negative, supporting its use as a conservative borrowing framework under globally null conditions.

5.5 Understanding the Limits of Information Borrowing

Across all six scenarios, no single analytical strategy was uniformly optimal. Complete pooling achieved the best performance when subgroup exchangeability was fully satisfied but deteriorated rapidly once heterogeneity emerged. Independent analysis remained robust under severe heterogeneity but sacrificed efficiency when subgroups were genuinely similar.

The HBM occupied an intermediate position between these two extremes. Its principal advantage was not maximal efficiency under any single scenario, but rather the ability to adapt borrowing strength according to the observed degree of between-subgroup variation. The simulation results demonstrate that hierarchical borrowing is most beneficial when subgroups are neither completely homogeneous nor completely unrelated. Within this intermediate region, the model captures meaningful efficiency gains while preserving reliable subgroup-specific inference, serving as a practical framework when prior exchangeability assumptions are highly uncertain.

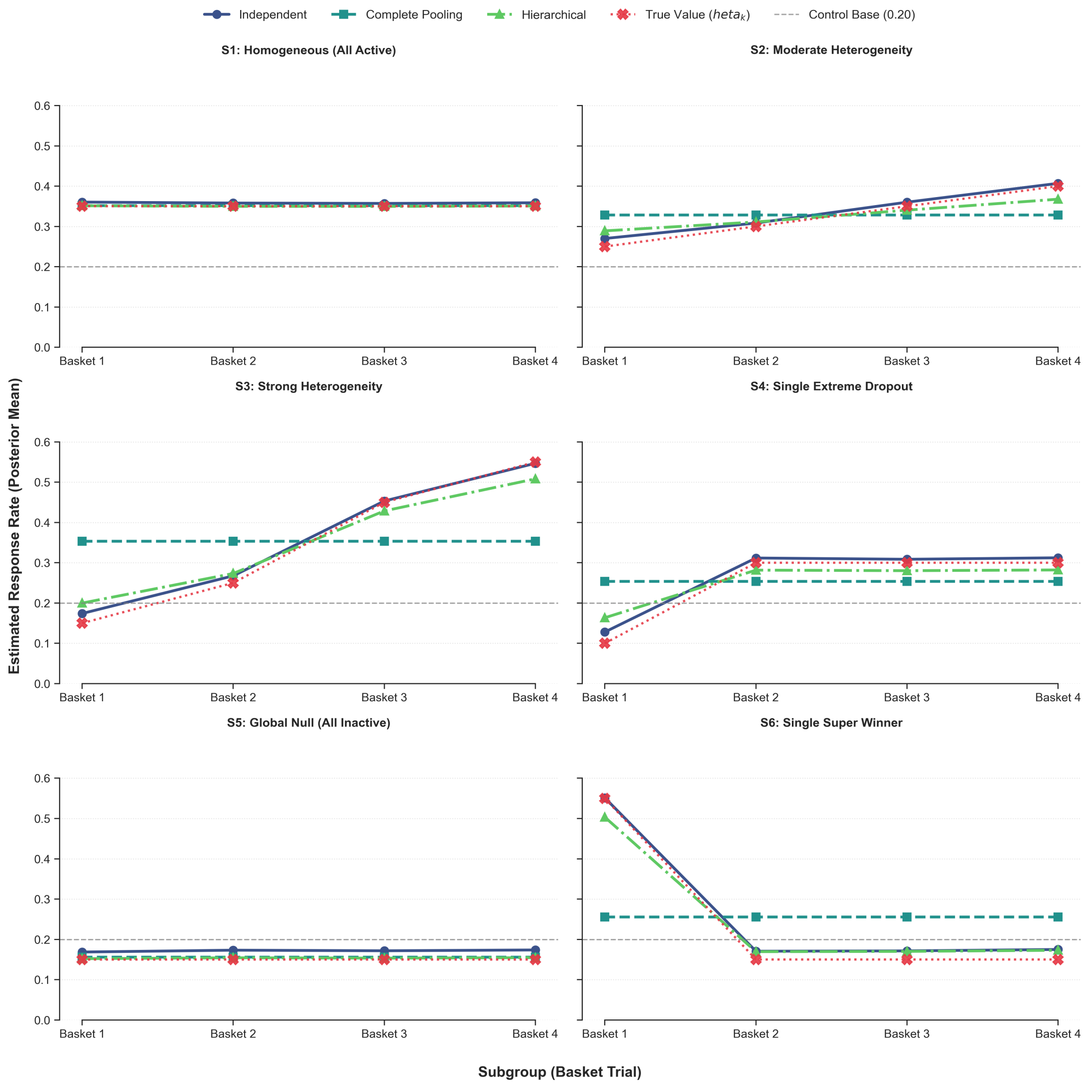

Figure 2: Posterior Mean Response Rate Estimates Across Six Basket Trial Scenarios. Three information borrowing frameworks are compared: independent analysis (no cross-basket borrowing), complete pooling (full borrowing), and hierarchical Bayesian modeling (adaptive borrowing). Red dotted lines indicate true underlying response rates for each basket; the grey dashed horizontal line represents the control baseline response rate of 0.20.

6 Discussion

This study evaluated the performance of Bayesian information borrowing across diverse subgroup heterogeneity scenarios to map its operational boundaries. While Berry (2025) emphasizes the flexibility of Bayesian designs, I argue that for regulatory purposes, the robustness of information borrowing is more important than maximum efficiency. A design that excels under idealized homogeneity but fails catastrophically under localized heterogeneity is not acceptable in regulatory settings. The simulation results validate this caution: complete pooling delivers optimal efficiency under perfect homogeneity yet collapses under non-exchangeability. By contrast, the HBM strikes a trade-off between efficiency and robustness by adaptively adjusting borrowing strength according to observed data.

The simulation also revealed an important limitation of this method. When groups are either fully similar or only slightly different, the HBM produced a higher Maximum Absolute Deviation than the fully pooled model. This indicates that with few cohorts ($K = 4$) and modest sample sizes ($N_k = 30$), the posterior distribution of the between-group variance parameter τ is highly uncertain. In unfavorable sampling allocations, this uncertainty will lead to larger estimation errors in certain subgroups. Simply put, the HBM's ability to adapt to the data comes at the cost of less stable estimates when compared to the fixed pooling approach.

Besides, several limitations should also be acknowledged. First, the simulations considered only binary outcomes with equal sample sizes across four subgroups, which simplifies the structure of confirmatory master protocols where treatment effects are evaluated against concurrent or external controls. Second, only one hierarchical specification was evaluated, and a formal prior sensitivity analysis for the heterogeneity parameter τ was omitted. Given that this hyperprior directly modulates borrowing velocity, evaluating alternative formulations remains critical in real-world analysis. Third, the study focused exclusively on horizontal borrowing across concurrent subgroups and did not consider longitudinal historical borrowing methods. Comparing these approaches within a common simulation framework would provide a broader understanding of the information borrowing landscape. Fourth, I attempted to access the publicly available data from the paper introduced by one of the guest speakers (Richeldi et al., 2022). However, the request was declined by the sponsor of the study, limiting the scope of this evaluation to simulated environments.

Future extensions should incorporate continuous and time-to-event endpoints, evaluate prior sensitivity over τ, and directly compare horizontal HBMs against historical control borrowing frameworks.

7 Conclusion

Clinical trials for rare diseases and small patient populations often face a primary challenge: there may simply not be enough participants to generate reliable evidence using traditional study designs. In such scenarios, innovative statistical methods provide a viable solution. Instead of treating every small group of patients as a separate experiment, researchers can look across related medical conditions or age groups to safely share information, thereby maximizing the value of every single patient's data.

By testing these approaches under a range of virtual conditions, this study evaluated how information-sharing methods perform when patient groups range from nearly identical to completely different. The results show that flexible, adaptive data-sharing methods provide the most practical path forward. They successfully boost the statistical power of a trial when patient groups behave similarly, while automatically limiting cross-group information sharing to avoid bias when meaningful differences emerge. In contrast, blindly pooling all patients into a single analysis may work well on paper but can lead to dangerously misleading conclusions if the groups are incompatible. These results suggest study teams planning trials with limited patient groups should consider flexible analysis strategies. To satisfy regulatory standards, any trial plan that uses data from related patient groups must include simulation assessments covering diverse clinical subgroup differences.

This report highlights that the role of modern biostatistics is not just about inventing sharper tools to extract more data from fewer people. Rather, as regulatory frameworks adapt to incorporate these advanced designs, the primary responsibility becomes determining exactly when and how such information can be shared without compromising safety. As clinical research increasingly focuses on narrowly defined patient groups, balancing scientific efficiency with patient safety and regulatory oversight remains essential to delivering effective treatments to the individuals who need them most.

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